



Solution Set

Question 1

A one-day loss L has values 0, 2, 7, and 15 million with probabilities 0.75, 0.15, 0.08, and 0.02. Compute $E[L]$, $\text{Var}(L)$, and the 98 percent VaR under the loss convention.

Answer key

The expectation is

$$E[L] = 0(0.75) + 2(0.15) + 7(0.08) + 15(0.02) = 1.16$$

The second moment is

$$E[L^2] = 0 + 4(0.15) + 49(0.08) + 225(0.02) = 9.02$$

Thus

$$\text{Var}(L) = 9.02 - 1.16^2 = 7.6744$$

The cumulative probabilities are 0.75, 0.90, 0.98, 1.00, so the 98 percent VaR is 7 million.

Question 2

Prove the inclusion-exclusion identity for two events and apply it to

$$P(A) = 0.4, \quad P(B) = 0.35, \quad P(A \cap B) = 0.12$$

Answer key

Adding the two event probabilities counts the overlap twice, so subtract the intersection once:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Numerically,

$$0.4 + 0.35 - 0.12 = 0.63$$

This identity is heavily used when combining loss events without double-counting joint scenarios.

Question 3

A rare jump loss has probability p and size J . Show why it can dominate expected loss even when p is small.

Answer key

If the loss is J with probability p and 0 otherwise,

$$E[L] = pJ$$

A small p can still produce a large contribution when J is very large. For $p=0.005$ and $J=500$ million, expected loss is 2.5 million. Tail severity matters as much as event probability.

Question 4

State the axioms of probability and explain which axiom rules out negative scenario probabilities.

Answer key

The core axioms are nonnegativity, total probability one, and countable additivity:

$$\mathbb{P}(A) \geq 0, \quad \mathbb{P}(\Omega) = 1, \quad \mathbb{P}\left(\bigcup_i A_i\right) = \sum_i \mathbb{P}(A_i) \text{ for disjoint } A_i$$

Nonnegativity rules out negative scenario probabilities. In risk systems, negative probabilities usually indicate an invalid calibration or numerical artifact.

Question 5

Define VaR as a quantile of a loss distribution and discuss what happens when the CDF has jumps.

Answer key

For confidence level alpha, a common definition is

$$\text{VaR}_\alpha(L) = \inf \{ \ell : F_L(\ell) \geq \alpha \}$$

If the CDF jumps, many losses may share probability mass. The infimum definition selects the smallest loss level whose cumulative probability reaches alpha. This matters for discrete scenario VaR.

Question 6

Let L be a nonnegative loss. Prove Markov's inequality and explain its risk interpretation.

Answer key

For $a > 0$,

$$\mathbb{E}[L] \geq \mathbb{E}[L \mathbf{1}_{L \geq a}] \geq a \mathbb{P}(L \geq a)$$

Therefore

$$\mathbb{P}(L \geq a) \leq \frac{\mathbb{E}[L]}{a}$$

The bound is often loose, but it shows that a finite expected loss limits how much probability can sit at extremely large losses.

Question 7

Two loss events are independent with probabilities 3 percent and 5 percent. Compute the probability of at least one event, and compare with the simple sum.

Answer key

Independence gives

$$\mathbb{P}(A \cap B) = 0.03(0.05) = 0.0015$$

Therefore

$$\mathbb{P}(A \cup B) = 0.03 + 0.05 - 0.0015 = 0.0785$$

The simple sum 0.08 double-counts the joint event, so it is slightly too high.

Question 8

Explain the difference between real-world probability and risk-neutral probability in one paragraph.

Answer key

Real-world probability describes actual frequencies or forecasting beliefs and is used for risk measurement, backtesting, and economic capital. Risk-neutral probability is a pricing device under which discounted tradable asset prices are martingales. A derivative price is an expectation under the risk-neutral measure, not generally under the real-world measure.

Question 9

A risk analyst estimates $P(L > 10) = 0.01$ from 250 observations. Why is this estimate statistically fragile?

Answer key

A 1 percent tail event corresponds to only 2.5 observations in a sample of 250. The estimate depends heavily on a tiny number of points, so sampling error is large. This is why interviewers expect caution around high-confidence historical VaR with short histories.

Question 10

Define a sigma-algebra and explain why it appears in conditional risk measurement.

Answer key

A sigma-algebra is a collection of events closed under complements and countable unions. It represents a consistent information set. Conditional expectations such as

$$\mathbb{E}[L_{t+1} \mid \mathcal{F}_t]$$

condition on the sigma-algebra $\mathcal{F}_t$, meaning all information observable at time t .

Question 11

If

$$X$$

is integrable, prove

$$\mathbb{E}[X] = \int_0^\infty \mathbb{P}(X > x) dx - \int_0^\infty \mathbb{P}(X < -x) dx$$

Answer key

Decompose X into positive and negative parts, $X = X^+ - X^-$. For a nonnegative random variable Y , the layer-cake identity gives

$$\mathbb{E}[Y] = \int_0^\infty \mathbb{P}(Y > x) dx$$

Apply this to X^+ and X^- :

$$\mathbb{E}[X] = \int_0^\infty \mathbb{P}(X > x) dx - \int_0^\infty \mathbb{P}(X < -x) dx$$

This identity connects expected loss to the whole tail curve.

Question 12

Why is expected shortfall considered more informative than VaR for tail risk?

Answer key

VaR identifies a quantile threshold but ignores how bad losses are beyond that threshold. Expected shortfall averages losses in the tail:

$$\text{ES}_\alpha(L) = \mathbb{E}[L \mid L \geq \text{VaR}_\alpha(L)]$$

with technical adjustments for discontinuities. It is more sensitive to tail severity and is subadditive under standard definitions.